

# Kun-Hee Son

[songunhee5426@gmail.com](mailto:songunhee5426@gmail.com) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

## RESEARCH INTEREST

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My research focuses on **language-grounded robot manipulation**, **imitation learning** (particularly Diffusion Policy and Action Chunking Transformer), and **sim-to-real transfer** for contact-rich tasks. I am especially interested in building robust, language-conditioned policies that bridge high-level instructions and low-level action execution on real manipulators.

## EDUCATION

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- **Kongju National University** March 2020 – February 2027 (Expected)  
*B.S. in Artificial Intelligence* *Gongju, Republic of Korea*
  - GPA: 4.15 / 4.50
- **University of Nevada, Las Vegas (UNLV), USA** January 2024  
*Exchange Program (Cultural & Academic)* *Las Vegas, USA*

## EXPERIENCE

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- **AI & Robotics Hackathon** March 2026  
*Developer* *Doosan 6-axis Robot Arm + ROS2*
  - Built an end-to-end **language-conditioned manipulation pipeline**: Speech (Whisper) → Language (Gemma LLM) → Object detection (YOLO) → 3D coordinates → Manipulation
  - Encountered the **spatial grounding ambiguity problem** (e.g., “place it to the left of the cup”), directly motivating my interest in LINGO-Space-style probabilistic grounding
  - Implemented camera-to-robot coordinate calibration
- **CES 2024 Attendee** January 2024  
*Las Vegas, USA*
- **NVIDIA Deep Learning Institute (DLI)** 2025  
*Certificate Holder*

## PROJECTS

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- **AI for Industry Challenge (AIC) — UR5e Connector Insertion** Feb 2026 – Present  
*Team Member | Data collection & model implementation | Intrinsic + Open Robotics*
  - **Data Collection**: Building and operating a Gazebo + ROS2 Kilted simulation environment for teleoperation data collection of contact-rich connector insertion
  - **Model Implementation**: Implementing and comparing **ACT (Action Chunking Transformer)**, **Diffusion Policy**, and  $\pi_0$ -**FAST** on the collected data
  - Direct experience with the *single-granularity action chunking* limitation that DiSPo (ICRA 2026) addresses
- **TurtleBot3-Based Autonomous Transport Robot** Jul 2025 – Sep 2025  
*Team Lead | Kongju National University*
  - Led development of an autonomous transport robot using TurtleBot3 with ROS2 and YOLO
  - **Sim-to-real deployment**: transferred Gazebo-trained policy to Raspberry Pi 5 real hardware
  - Implemented automatic charging navigation and real-time web dashboard
- **Vehicle Trajectory Generation from Aerial Drone Video** Jan 2025 – Jun 2025  
*Developer | Kongju National University*

- Developed a CVAE model to generate realistic vehicle trajectories from YOLO-tracked drone footage
- Addressed mode collapse through preprocessing pipeline (interpolation, length filtering)

- **Curriculum Learning in Image Classification** Jan 2024 – May 2024  
*Researcher | Kongju National University*

- Investigated curriculum learning strategies on CCT under label noise (CIFAR-10)
- Published at 2024 KICS Spring Conference

- **Autonomous Driving Competition (DSCU)** July 2023  
*Developer | Kongju National University*

- Implemented traffic light recognition (OpenCV) and object detection (YOLO) on a 1/10 scale RC car
- Won Grand Prize (1st Place)

### PUBLICATIONS

D=DOMESTIC CONFERENCE

- [D.1] K.-H. Son. **Performance Comparison of Curriculum Learning Priority Strategies in Image Classification.** 2024 KICS Spring Conference, Korea Institute of Communication Sciences, 2024.
- [D.2] K.-H. Son. **Research on Adversarial AI Technology for Autonomous Ship Protection.** Korea Information Processing Society, 2024.
- [D.3] K.-H. Son. **Improving Collision Avoidance Performance of Reinforcement Learning Agents for Autonomous Ships.** 2024.

### RESEARCH SURVEY (SELF-DIRECTED)

- **Imitation Learning for Manipulation:** ACT, Diffusion Policy,  $\pi_0$ , Flow Matching — focus on action head design and inference efficiency
- **Language Grounding:** VLA architectures (OpenVLA, PaliGemma), probabilistic space grounding, language-conditioned policies
- **Sim-to-Real Manipulation:** domain randomization, teacher–student distillation, representation learning for deformable objects
- **VTLA Survey:** GelSight, DIGIT, Touch100k, TVL datasets, tactile-visual fusion methods

### AWARDS & HONORS

- Grand Prize, Big Data / AI Hackathon 2024
- Grand Chancellor’s Award, KNU Generative AI Content Contest 2024
- Grand Prize (Excellence), SW-Centered University Essay Contest 2024
- Merit Award (x2), KNU Entrepreneurship Contest & AWS Hackathon 2024
- Grand Prize (1st Place), DSCU Autonomous Driving Competition 2023
- Excellence Award, DSCU Autonomous Driving Hackathon 2023
- CEO Award, Electronic Times Mobility Creative S&T Contest 2023

## SKILLS

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- **Languages:** Korean: Native, English: TOEIC Speaking IM3 (130)
- **Programming Languages:** Python, C/C++, PyTorch, TensorFlow
- **Robot Systems:** ROS2 (Humble/Jazzy/Kilted), MoveIt2, Gazebo
- **Tools:** Docker, Git/GitHub, Linux (Ubuntu), AWS, FastAPI
- **Hardware:** TurtleBot3, Doosan Robot Arm, UR5e, Raspberry Pi 5